📊 Retail Sales Intelligence Dashboard

Power BI Project Report

🔷 1. Introduction

In the modern retail industry, data plays a critical role in driving strategic decisions. This project focuses on developing an **interactive Retail Sales Intelligence Dashboard** using Microsoft Power BI.

The dashboard transforms raw sales data into meaningful insights, enabling stakeholders to monitor performance, identify trends, and optimize business strategies.

🎯 2. Objectives

The key objectives of this project are:

✓ Analyze retail sales data effectively
✓ Build an interactive dashboard using Power BI
✓ Track key business KPIs
✓ Identify top-performing products and categories
✓ Enable data-driven decision-making

🛠️ 3. Tools & Technologies

Tool	Purpose
Power BI Desktop	Data modeling & visualization
Power BI Service	Dashboard publishing
DAX	Creating calculated measures
Data Modeling	Star schema design

🗂️ 4. Dataset Overview

The dataset contains retail business data including:

- 📦 Product Name

- 🏷️ Category & Sub-Category
- 💰 Sales & Profit
- 📊 Quantity & Orders
- 🌍 Region & State
- 🗓️ Monthly Sales Data

🧩 5. Data Modeling (Star Schema)

A **star schema architecture** was implemented for better performance and scalability.

🔹 **Dimension Tables:**

- Dim_Category
- Date (Month/Year)
- Region

🔹 **Fact Table:**

- Fact_ProductSales

🔹 **Relationships:**

- Dim_Category → Fact_ProductSales
- Date → Monthly_Sales
- Region → Region_Sales

✓ All relationships use **single-direction filtering**
✓ Avoided many-to-many relationships

📐 6. Key Metrics (DAX Measures)

Metric	Description
Total Revenue	Sum of all sales
Total Profit	Sum of profit
Average Order Value	Revenue per order
Profit Margin %	Profitability Ratio
Total Orders	Number of transactions

📊 7. Dashboard Design

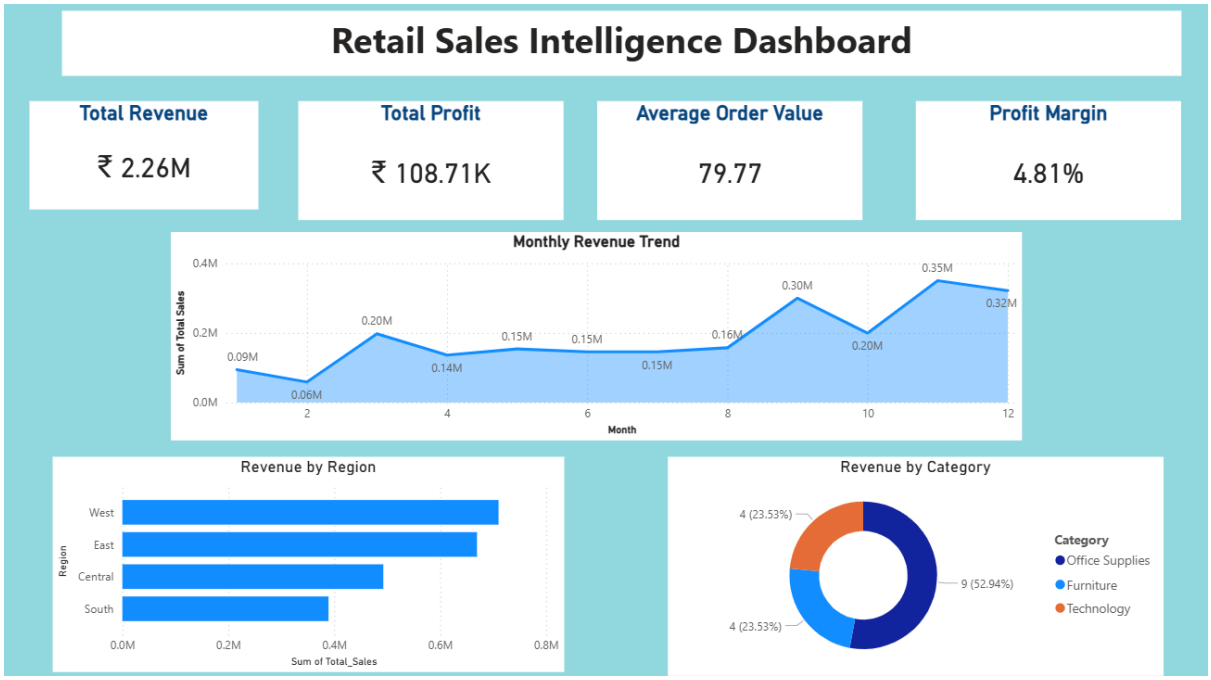
🟦 Page 1: Executive Overview

✓ KPI Cards

- Total Revenue
- Total Profit
- Average Order Value
- Profit Margin

✓ Visuals:

- 📈 Monthly Revenue Trend
- 📊 Revenue by Region
- 🍩 Revenue by Category

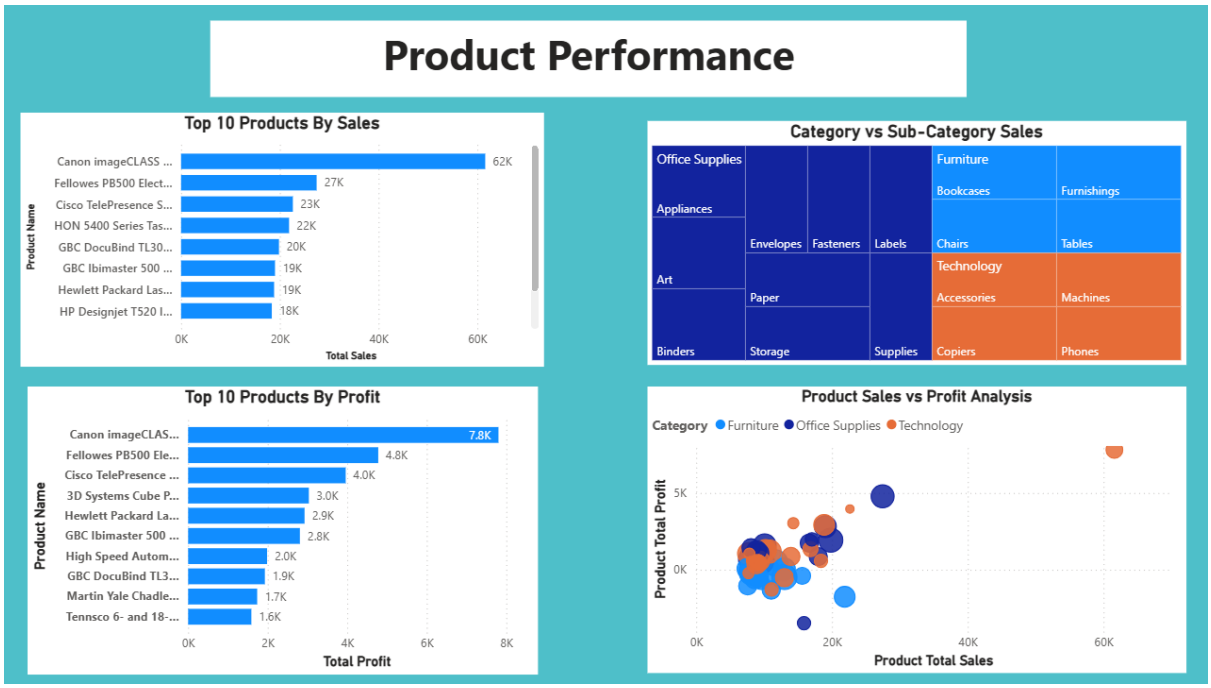


🟩 Page 2: Product Performance

✓ Visuals:

- 🏆 Top 10 Products by Sales
- 💹 Top 10 Products by Profit

- 🧱 Category vs Sub-Category (Treemap)
- ⚖️ Sales vs Profit (Scatter Plot)



🔍 8. Key Insights

- High sales products do not always generate high profit
- Technology category shows strong profitability
- Regional performance varies significantly
- Seasonal trends observed in monthly sales

⚠️ 9. Challenges Faced

- Managing multiple tables and relationships
- Designing an optimized data model
- Avoiding filter ambiguity
- Maintaining visual consistency

✅ 10. Solutions Implemented

✓ Implemented star schema
✓ Used dimension tables for filtering

✓ Applied single-direction relationships
✓ Cleaned model by hiding unnecessary tables
✓ Standardized dashboard layout

📄 11. Conclusion

The dashboard successfully converts raw retail data into actionable insights. It provides a clear understanding of business performance and supports effective decision-making.

This project demonstrates strong capabilities in:

✓ Data Analysis
✓ Data Modeling
✓ Dashboard Design
✓ Business Intelligence

🚀 12. Future Enhancements

- Real-time data integration
- Predictive analytics
- Drill-through reports
- Advanced slicers
- Cloud-based sharing

📚 13. References

- Microsoft Power BI Documentation
- DAX Guide
- Data Visualization Best Practices